

Physics-Informed Candidate Features for Short-Window Ride Events

I. PURPOSE AND SCOPE

This document defines a *candidate feature space*; it does not prescribe one universally optimal KPI per sensor channel. The ride-analysis review in [1] explicitly states that KPI suitability depends on the road profile, manoeuvre, measured quantity, and subjective attribute. In particular, isolated obstacles are commonly described by peak and peak-to-peak quantities, long road profiles by RMS-based quantities, and strongly transient vibration by running-RMS or fourth-power quantities.

ISO 2631-1 and BS 6841 define ways to quantify human exposure to whole-body vibration [2], [3]. They provide physically meaningful frequency weightings, but they do not establish which feature best predicts the binary feedback in this data set. The ISO-based quantities below are therefore used as physics-informed candidates. Their predictive relevance must be determined using held-out-driver posterior-predictive evaluation.

II. MEASURED AND DERIVED QUANTITIES

Let $x(t)$ denote a signal over a window of duration T and sampling rate f_s . The basic descriptors used by the feature pipeline are

$$\begin{aligned} \text{RMS}(x) &= \sqrt{\frac{1}{T} \int_0^T x^2(t) dt}, \\ \text{AbsPeak}(x) &= \max_t |x(t)|, \\ \text{P2P}(x) &= \max_t x(t) - \min_t x(t), \\ \text{CF}_{\text{raw}}(x) &= \frac{\text{AbsPeak}(x)}{\text{RMS}(x)}. \end{aligned}$$

The code names are `rms`, `abs_peak`, `p2p`, and `crest`, respectively. These are generic signal descriptors. In particular, `CFraw` is not the standards-based weighted crest factor defined below.

For a measured rate signal, differentiation produces the quantity used by the review's acceleration-based KPIs:

$$\dot{x}(t) = \frac{dx(t)}{dt}.$$

Thus, differentiating pitch, roll, or yaw rate gives angular acceleration, while differentiating heave velocity gives vertical acceleration. The corresponding code names are `rms_deriv`, `p2p_deriv`, `abs_peak_deriv`, and `vdv_deriv`. Differentiation amplifies sensor noise, so smoothing, sampling rate, and finite-difference accuracy must be treated as part of the feature definition.

TABLE I
ISO 2631-1 WEIGHTING PARAMETERS USED BY THE CURRENT CHANNELS.

Weight	f_1	f_2	$f_3 = f_4$	Q_4	Scale step
W_k	0.4	100	12.5	0.63	$f_5 = 2.37, f_6 = 3.35, Q_5 = Q_6 = 0.91$
W_d	0.4	100	2.0	0.63	none
W_e	0.4	100	1.0	0.63	none

III. ISO FREQUENCY-WEIGHTED QUANTITIES

For a translational or rotational acceleration $a(t)$, define

$$a_w(t) = \mathcal{F}^{-1}\{W(f) \mathcal{F}\{a\}(f)\}.$$

With $s = j2\pi f$ and $\omega_i = 2\pi f_i$, the ISO weighting is constructed from

$$\begin{aligned} H_h(s) &= \frac{s^2}{s^2 + \sqrt{2}\omega_1 s + \omega_1^2}, \\ H_\ell(s) &= \frac{\omega_2^2}{s^2 + \sqrt{2}\omega_2 s + \omega_2^2}, \\ H_t(s) &= \frac{1 + s/\omega_3}{1 + s/(Q_4\omega_4) + (s/\omega_4)^2}, \\ H_s(s) &= \left(\frac{\omega_5}{\omega_6}\right)^2 \frac{1 + s/(Q_5\omega_5) + (s/\omega_5)^2}{1 + s/(Q_6\omega_6) + (s/\omega_6)^2}. \end{aligned}$$

The scale-step normalization in H_s is $(\omega_5/\omega_6)^2$. Reversing this ratio preserves the frequency-response shape but multiplies W_k by approximately four, so it is not the ISO-normalized physical quantity.

The relevant parameter sets are summarized in Table I. The resulting magnitudes are

$$\begin{aligned} W_k &= |H_h H_\ell H_t H_s|, \\ W_d &= |H_h H_\ell H_t|, \\ W_e &= |H_h H_\ell H_t|. \end{aligned}$$

where the transition parameters differ as shown in Table I. Their intended directions are

- W_k : vertical translational acceleration (z direction),
- W_d : longitudinal and lateral translational acceleration (x, y directions),
- W_e : rotational acceleration about the roll, pitch, and yaw axes.

TABLE II
INTERPRETATION OF EXISTING FEATURE-PIPELINE NAMES.

Code name	Interpretation
<code>wrms_z</code> , <code>wrms_xy</code>	frequency-weighted translational RMS
<code>wrms_rot</code> , <code>wrms_z_d</code>	differentiate rate, then weighted RMS
<code>crest</code>	unweighted generic peak-to-RMS ratio
<code>vdv</code> , <code>vdv_deriv</code>	unweighted fourth-power feature
<code>mtvv</code>	unweighted running-RMS maximum

The standards-based weighted RMS and transient quantities are

$$\begin{aligned}
 \text{WRMS}(a; W) &= \sqrt{\frac{1}{T} \int_0^T a_w^2(t) dt}, \\
 \text{CF}_w(a; W) &= \frac{\max_t |a_w(t)|}{\text{WRMS}(a; W)}, \\
 \text{RWRMS}(t_0) &= \sqrt{\frac{1}{\tau_0} \int_{t_0-\tau_0}^{t_0} a_w^2(t) dt}, \\
 \text{MTVV}(a; W) &= \max_{t_0} \text{RWRMS}(t_0), \\
 \text{VDV}(a; W) &= \left(\int_0^T |a_w(t)|^4 dt \right)^{1/4}.
 \end{aligned}$$

The review uses $\text{CF}_w = 6$ as a BS 6841-based stationary/transient guideline, whereas ISO 2631-1 normally regards the basic weighted-RMS method as sufficient up to a crest factor of 9. Neither threshold should be applied to the current unweighted `crest` feature.

A. Current implementation status

The last three rows are useful exploratory statistics, but they must not be reported as ISO CF, VDV, or MTVV unless weighting is applied before their calculation.

IV. SIGNAL-TO-CANDIDATE MAPPING

Table III maps each available signal to physically meaningful candidates. The ordering distinguishes event-oriented base candidates from standards-inspired or exploratory additions; it is not a claim of predictive importance.

V. SELECTION BY EVENT REGIME

For the current short windows around a reported event, the isolated-event row is the most appropriate starting point. WRMS remains a useful perceptually weighted candidate, but it should compete with peak, P2P, and RMS features rather than replace them. A four-second window also provides limited low-frequency resolution; the resulting WRMS is better described as an ISO-inspired relative feature than as a standards-compliant exposure measurement.

TABLE III
CANDIDATE METRICS FOR THE AVAILABLE CHANNELS. NO ROW DEFINES A UNIVERSAL PRIMARY KPI.

Signal	Physical quantity	Event-oriented candidates	Additional candidates	Applicability / caveat
IMU_LongAccelVal	longitudinal body acceleration a_x	abs_peak, p2p, rms	wrms_xy, p95_abs, vdv, crest	impact and traction/braking response; W_d is directionally valid
IMU_VerAccelVal	vertical body acceleration a_z	abs_peak, p2p, rms	wrms_z, p95_abs, vdv, crest, mtvv	central ride signal; z_{nod} is not valid on acceleration
Pitch_rate_6D	pitch rate $\dot{\theta}$; derive θ	abs_peak_deriv, p2p_deriv, rms_deriv	wrms_rot, vdv_deriv, raw-rate rms	RMS/P2P angular acceleration is supported for pitch control and braking
Roll_rate_6D	roll rate $\dot{\phi}$; derive ϕ	abs_peak_deriv, p2p_deriv, rms_deriv	wrms_rot, vdv_deriv, raw-rate rms	relevant to asymmetric roads, road copying, and cornering
IMU_YawRtVal	yaw rate $\dot{\psi}$; derive ψ	p2p_deriv, rms_deriv	wrms_rot, raw-rate rms, crest	mathematically valid rotation; mainly a handling/context candidate
Bounce_rate_6D	heave velocity $\dot{z}$; derive z	abs_peak_deriv, p2p_deriv, rms_deriv	wrms_z_d, vdv_deriv, raw-rate rms	derived vertical acceleration is valid; displacement-decay KPIs are not
IMU_LatAccelVal	lateral body acceleration a_y	abs_peak, p2p, rms	wrms_xy, p95_abs, vdv, crest	most relevant to cornering or asymmetric events; W_d is directionally valid
IMU_RollRtVal	IMU roll rate $\dot{\phi}$; derive ϕ	same as Roll_rate_6D	use as an alternative sensor/source	likely redundant with Roll_rate_6D; compare rather than assume both

TABLE IV
THE EVENT REGIME DETERMINES WHICH STATISTICS ARE DEFENSIBLE CANDIDATES.

Regime / target attribute	First candidates	Conditional additions
Isolated bump, cleat, or short impact	absolute peak, P2P, RMS of linear/angular acceleration	weighted RMS; weighted RWRMS/MTVV/VDV for strong transients
Long, flat, rough, or approximately stationary road	RMS and frequency-weighted RMS	band-limited energy, vibration total value
Acceleration or braking pitch	RMS/P2P pitch angular acceleration, longitudinal acceleration	pitch settling/response time when a steady state is identifiable
Cornering or asymmetric road input	RMS/P2P roll angular acceleration, lateral acceleration	yaw and steering/context quantities
Oscillation decay after a step	displacement settling time, overshoot, third displacement peak	only when displacement and a defensible steady state are available
Road holding	suspension deflection, suspension working space, dynamic wheel load	unavailable from the current selected IMU channels alone

VI. EVIDENCE-BASED PRIORITIES FOR THE CURRENT BUMP TASK

Table V translates the review’s event- and attribute-specific evidence into priorities for the available signals. “First” means that the review directly supports the same physical response for an isolated impact or a closely related body-control attribute. “Conditional” means that the statistic is supported only for a particular event type, requires an additional diagnostic, or is a physically reasonable extrapolation without a direct pairing in Table 4 of [1].

TABLE V
EVIDENCE-BASED CANDIDATE PRIORITY FOR THE CURRENT SHORT-WINDOW BUMP TASK.

Signal	First candidates	Conditional candidates	Evidence and rationale
IMU_LongAccelVal	p2p	abs_peak, rms, wrms_xy	Figure 11 reports a direct relationship between longitudinal seat-rail P2P and bump impact harshness. The current CG-level IMU is a proxy for that measurement location.
IMU_VerAccelVal	p2p, abs_peak, rms	wrms_z, p95_abs, properly weighted MTVV/VDV	Section 2.2 identifies vertical peak and P2P for isolated obstacles. Table 4 pairs bump impact with seat vertical P2P and pairs vertical/bounce control with vertical RMS acceleration.
Pitch_rate_6D	rms_deriv, p2p_deriv	abs_peak_deriv, wrms_rot, settling/response time	Table 4 pairs RMS pitch angular acceleration with pitch control and P2P/RMS pitch angular acceleration with pitch under acceleration or braking. Differentiation maps the measured rate to this quantity.
Roll_rate_6D, IMU_RollRtVal	rms_deriv	p2p_deriv for asymmetric impacts; abs_peak_deriv, wrms_rot	Table 4 pairs RMS roll angular acceleration with road copying. Section 2.2 also permits P2P on angular body acceleration for isolated events. The two roll channels should be compared as alternative sources.
Bounce_rate_6D	rms_deriv, p2p_deriv, abs_peak_deriv	wrms_z_d, properly weighted transient metrics, raw-rate rms	Differentiation produces heave acceleration. Table 4 directly pairs RMS sprung-mass vertical acceleration with vertical ride/bounce control, while Section 2.2 supports peak/P2P vertical acceleration for isolated obstacles.
IMU_LatAccelVal	none for a generic symmetric bump	p2p, rms, abs_peak for asymmetric impacts; wrms_xy for cornering/repeated vibration	The review associates lateral acceleration with choppiness and cornering, but Table 4 gives no direct generic-bump pairing. Its relevance depends on the event geometry.
IMU_YawRtVal	none for a generic bump	raw-rate rms, rms_deriv, p2p_deriv, wrms_rot for directional/asymmetric events	Table 4 contains no direct yaw-to-bump or yaw-to-ride-attribute pairing. W_e is mathematically valid for yaw angular acceleration, but predictive relevance is a handling/context hypothesis.

A. Evidence hierarchy and scope

The review warns that the discussed KPIs are “not universally applicable to every type of manoeuvre and road profile” [1]. Accordingly, evidence is treated in the following order:

- 1) a metric paired with the same manoeuvre and subjective attribute in Table 4;
- 2) a metric explicitly recommended for the same physical response in Section 2.2;
- 3) a metric shown in the review’s correlation examples, especially Figure 11;
- 4) a directionally or dimensionally valid extrapolation that still requires empirical testing.

The review concerns attribute-specific ride assessment, whereas the current target is binary feedback on short bump events. It therefore supplies a defensible prior over candidates, not a guarantee that any one metric predicts the current label.

B. Vertical impact and heave response

Section 2.2 calls vertical absolute peak and P2P the “most frequently selected indicators ... during isolated events” [1]. Table 4 makes the distinction more specific: P2P seat vertical

acceleration is paired with small and large impact on a bump, whereas RMS seat vertical acceleration is paired with the corresponding impact on a cleat. The same table pairs RMS sprung-mass vertical acceleration with vertical ride control or bounce on bumpy roads with high wavelengths.

Consequently, p2p and abs_peak have the strongest isolated-impact basis for IMU_VerAccelVal, while rms captures sustained body response and remains a first candidate rather than a fallback. For Bounce_rate_6D, differentiation produces the heave acceleration used by the paper. This makes rms_deriv directly relevant to bounce control, with p2p_deriv and abs_peak_deriv capturing the transient impact. The current centre-of-gravity measurements remain proxies for the seat and sprung-mass locations used in the cited correlations.

C. Pitch and roll body control

Table 4 directly pairs RMS pitch angular acceleration with pitch control on flat and uneven roads. It also lists settling time, P2P pitch angular acceleration, and RMS pitch angular acceleration for pitch under acceleration or braking. The

isolated-event discussion in Section 2.2 states that P2P can also be evaluated on pitch and roll angular body acceleration.

For the rate channels, the matching quantities are obtained only after differentiation. Hence `rms_deriv` and `p2p_deriv` are the first pitch candidates. For roll, Table 4’s direct evidence is RMS roll angular acceleration for road copying, making `rms_deriv` the first general candidate. `p2p_deriv` becomes especially relevant for a one-wheel or otherwise asymmetric bump. Absolute angular-acceleration peak and W_e -weighted RMS are defensible additions, but the review does not establish them as the primary correlation for these attributes.

D. Longitudinal impact response

Table 4 focuses its small- and large-impact rows on vertical seat acceleration. The direct longitudinal evidence appears instead in Figure 11: front- and rear-seat-rail longitudinal P2P are shown in correlation with impact harshness for a speed bump. This supports `p2p` as the first longitudinal candidate. Absolute peak, RMS, and W_d -weighted RMS remain plausible additions, but their bump-specific subjective correlation is less direct in this review. Sensor-location differences again prevent interpreting the CG IMU feature as an exact replica of the seat-rail KPI.

E. Lateral and yaw responses

The review relates lateral body acceleration to choppiness and discusses cornering manoeuvres, while roll is used for road-copying and lane-change behaviour. It does not provide a Table 4 pairing between lateral acceleration or yaw and generic bump impact. These signals should therefore have no unconditional first statistic for a nominally symmetric bump. They become meaningful candidates when the data contain asymmetric road input, single-wheel impact, cornering, or a directional correction after the obstacle. The condition is on the event, not merely on the mathematical availability of a statistic.

F. Why weighted and dose metrics are conditional

The review lists WRMS as a ride-comfort KPI used for bumps as well as longer road profiles, so it remains a legitimate candidate. However, its most direct bump-to-subjective-attribute examples use P2P or RMS rather than WRMS. On a four-second event window, WRMS is also a short-window frequency-shaped summary rather than a representative whole-body exposure measurement.

RWRMS, MTVV, and VDV are motivated when ordinary RMS can underestimate occasional shocks or strongly transient vibration. Their priority should therefore depend on a correctly computed weighted crest factor or on a predefined transient-analysis design. The current raw `crest`, `mtvv`, and `vdv` implementations can be explored as generic shape and dose features, but they do not inherit the standards-based justification without frequency weighting.

G. Resulting interpretation

For the present task, the paper supports a more precise statement than “use peak statistics for every signal”: P2P and absolute peak are first candidates for impact severity on the vertical and longitudinal response, while RMS acceleration is a first candidate for heave, pitch, and roll body control. P2P angular acceleration is first or near-first when the bump induces a distinct pitch/roll transient. Lateral, yaw, frequency-weighted, and dose-based features are conditional candidates whose value depends on event geometry, spectral content, and the target driver’s response.

VII. UNAVAILABLE OR CONDITIONALLY VALID METRICS

A. Third positive peak

The review defines $z_{s,\text{nod}}$ as the third positive peak of the *sprung-mass vertical displacement* $z_s(t)$ after an obstacle. Applying this operator directly to vertical acceleration does not measure the same oscillation-decay property. It should be excluded unless displacement is measured or reconstructed with a validated, drift-controlled estimator.

B. Road-holding metrics

The root integral of suspension-deflection PSD σ_{SD} , RMS suspension deflection, suspension working space, and normalized dynamic wheel-load RMS require suspension travel or vertical tyre-load signals. They must not be computed from body acceleration merely because the same mathematical operation is available. Candidate raw channels for these quantities must first be assigned verified physical meanings and units.

VIII. MEASUREMENT AND PREPROCESSING LIMITATIONS

- ISO whole-body comfort evaluation is defined at body-contact locations such as the seat surface, seat back, and feet. A 6DoF IMU near the vehicle centre of gravity is a useful vehicle-response sensor, but its weighted features are proxies for occupant exposure.
- Rotational acceleration must be expressed in rad/s^2 for physical ISO reporting. A constant degree-to-radian conversion is immaterial after z-scoring, but it is material for reported KPI values.
- Per-axis WRMS values are not automatically comparable across directions. ISO total vibration combines orthogonal components using direction-dependent multiplying factors. Separate feature standardization intentionally removes this physical magnitude comparison.
- The current 10-Hz smoothing cutoff suppresses much of the review’s 10–25-Hz secondary-ride/shake band. Cutoff and downsampling choices must be included in sensitivity analysis if secondary ride is part of the target.
- Short-window FFT weighting is affected by spectral resolution, window boundaries, and leakage. The pre-processing and window duration are therefore part of the feature and must remain identical across training and evaluation folds.

IX. MODEL-BASED FEATURE SELECTION PRINCIPLE

Let $g = (\text{channel}, \text{statistic})$ be the atomic candidate and let S be a candidate subset. Selection should be based on the hierarchical model's predictive objective, not on the literature ordering or raw coefficient magnitude. For each held-out training driver u , fit the population posterior on the remaining drivers and evaluate

$$Q_t(S) = \frac{1}{U} \sum_u \frac{1}{|H_u|} \sum_{i \in H_u} \log p(y_{ui} \mid \mathcal{D}_{-u}, C_{u,t}, S),$$

where $t = 0$ measures cold-start transfer and $t > 0$ measures personalization after a fixed context budget. The predictive importance of group g is the paired difference

$$\Delta_g = Q(S) - Q(S \setminus g).$$

Channel-level importance removes all statistics belonging to a channel; statistic-level importance removes one pair. Final selection should choose the smallest subset within one standard error of the best macro-averaged held-out-driver score. The designated test driver must remain untouched until this procedure is complete.

X. CONCLUSION

The review and standards determine which candidates are physically defensible, but they do not determine the predictive winner for personalized binary feedback. For short bump-like events, peak, P2P, and RMS acceleration features are first-class candidates; weighted RMS and transient dose features provide complementary frequency and peak sensitivity. Feature importance must ultimately be reported separately for population-common effects, between-driver heterogeneity, and held-out-driver predictive gain.

REFERENCES

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